Database Schema

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Messenger Application Database Schema

Entities:

both could act as primary keys.

user (phone_num, name, username, password, IP, Port)
- ↳ 1st person perspective
- ↳ can be uniquely identified ⇒ essentially sender
- ↳ can send a message

contact (phone_num, name, username, IP, Port)
- ↳ the recipients, the senders will refer to.
- ↳ can be uniquely identified
- ↳ message will be sent to.

message(id, date, time, content)
- ↳ each message will have a unique id (by which identified)
- ↳ sender/recipient (client/server)
- ↳ at whose time
- ↳ what is sent

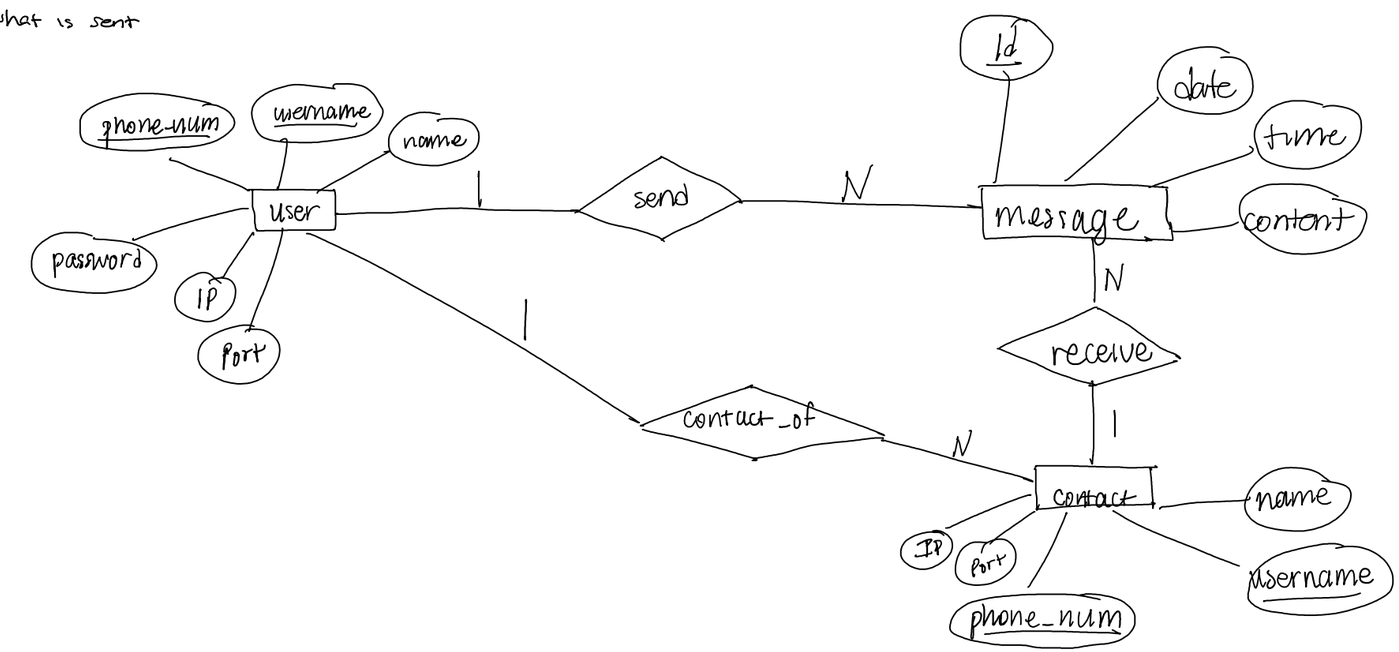
Relationships:

send (sender_id, message_id)
- ↳ the user (sender_id) sends a message (message_id)
- ↳ sender_id is a phone number
- ↳ message_id is an encoded id for the message

} a contact can send many messages

receive (recipient_id, recipient_id)
- ↳ the recipient (recipient_id) recieves a message sent by the sender
- ↳ recipient_id is a phone number
- ↳ message_id is an encoded id for message

} a contact can receive many messages

contact_of (sender_id, recipient_id)
- ↳ connects the sender & recipient
- ↳ both ids are phone_nums.

} a user can have many contacts